What do you want to learn?

ALICE L WALTON SCHOOL OF MEDICINE

Cyanotic Congenital Heart Defects (Right–to–Left Shunts)

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Adapted from: Cyanotic Obstructive Congenital Heart Defects

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Learning Objectives (1) Versions

After completing this brick, you will be able to:

1. Compare and contrast the clinical presentation, pathophysiology, diagnosis, and treatment of the primary types of cyanotic congenital heart defects.

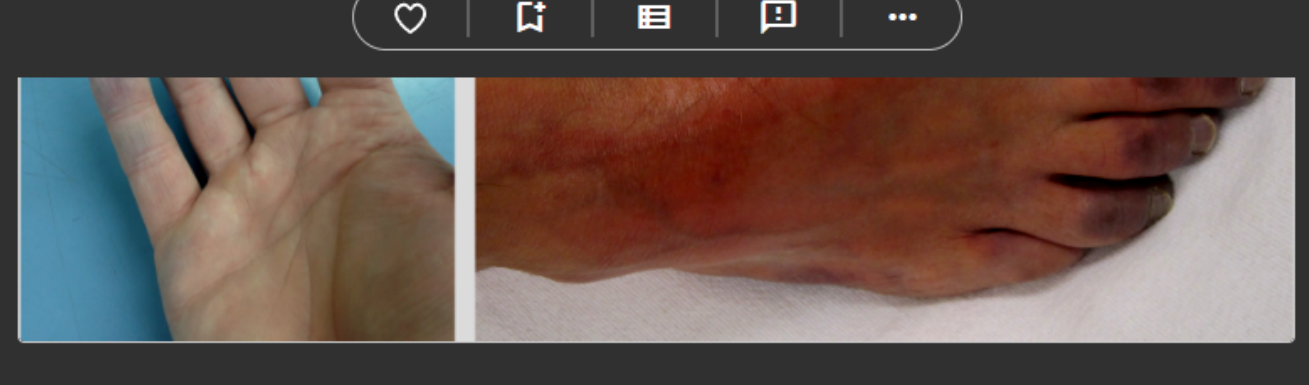


Figure 1 Cyanosis in nailbeds, fingers, and toes

Tetralogy of Fallot (TOF) and tricuspid atresia (TA) are two of the five most common cyanotic CHDs. They both also include **obstructive defects on the right side of the heart**. These obstructions limit blood flow into the pulmonary vasculature.

The other three common cyanotic CHDs are considered **"mixing defects,"** which are a subset of cyanotic CHDs where oxygen-rich and oxygen-poor blood mix, resulting in reduced oxygen delivery to the body and cyanosis. These CHDs are: truncus arteriosus, transposition of the great arteries, and total anomalous pulmonary venous return.

What Is Tetralogy of Fallot? (LO#1)

Tetralogy of Fallot is the name for the combination of four ("tetra") specific CHDs (Figure 2):

- Pulmonary valve stenosis
- Right ventricular (RV) hypertrophy (due to the valve stenosis)
- Ventricular septal
- Overriding aorta, right and receives

As mentioned, the primary significant defect in TOF is an obstruction, specifically **pulmonary valve stenosis**, which is a form of RV outflow tract (RVOT) obstruction. The degree of this obstruction will contribute to how and when a child develops symptoms of TOF. In severely restricted pulmonary artery flow, young neonates will develop cyanosis and require surgical intervention very early in life.

CLINICAL CORRELATION

Pulmonary valve stenosis is a CHD that can occur as part of TOF, but it more commonly exists in isolation. The symptoms depend on how severe the obstruction is, but commonly include an **ejection systolic, crescendo-decrescendo murmur** at the left upper sternal border. Severe isolated stenosis can cause right heart failure with dyspnea on exertion.

Children with less significant obstruction may present with **tet spells** (episodes of cyanosis, rapid breathing, and irritability). These typically happen when the RVOT obstruction suddenly worsens, which can occur for various reasons including agitation, feeding, or illness (fever, anemia, hypovolemia). Tet spells are unique to TOF, and a child having one should receive emergent intervention and further investigation.

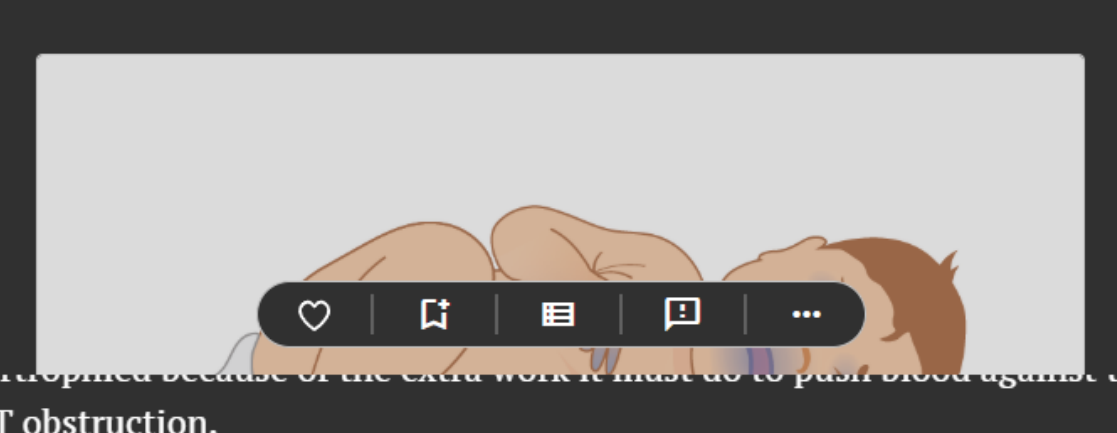
Pathophysiology

Mnemonic

crest cells.

Why does deoxygenated blood from the RV not mix with oxygenated blood in the LV in a newborn with an isolated VSD?

Complications. What accounts for tet spells? One common explanation is that during exertion (e.g., feeding or crying), the systemic arteries dilate, lowering systemic resistance. This suddenly lowered resistance will lower LV pressure and cause more deoxygenated blood to cross right to left across the VSD, leading to clinical tet spells (Figure 3).

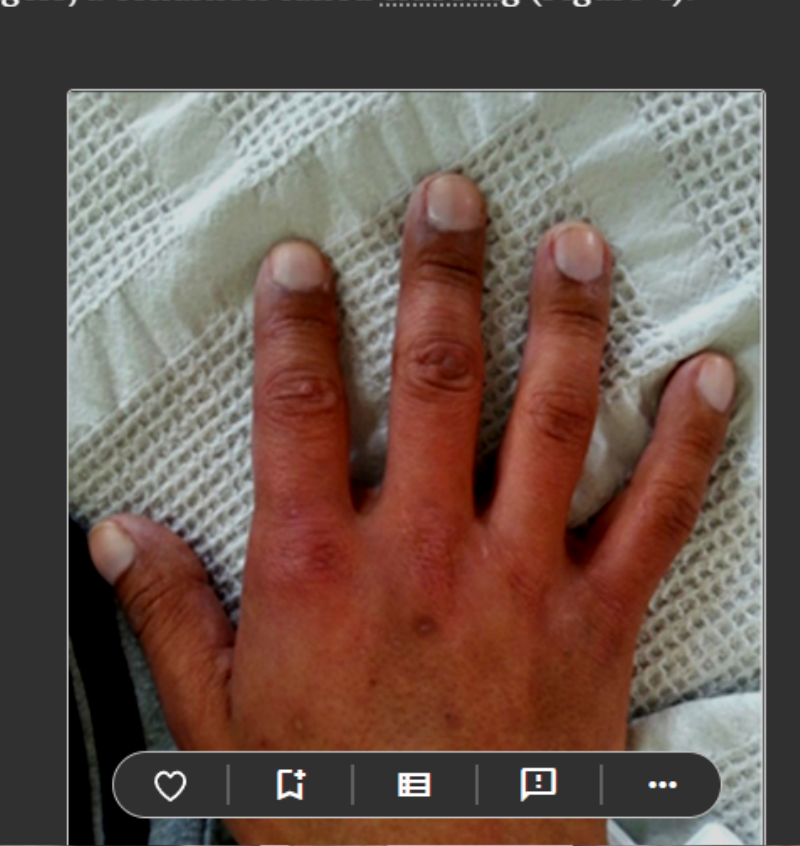


RVOT obstruction.

On auscultation, the patient may only have a **single S_2** from the normal aortic valve, as the pulmonic component is usually not heard because of pulmonary stenosis. This would most likely be detected by auscultating the heart during inspiration, when a normal physiologic splitting of S_2 would be expected but, in this case, would not occur.

A systolic ejection murmur can be heard at the upper left sternal border. The turbulent blood flow through the RVOT obstruction causes this, not the VSD, since flow through the VSD is not turbulent.

Over time, in addition to cyanosis, patients may develop thickening of the end part of the fingers, a condition called clubbing (Figure 4).



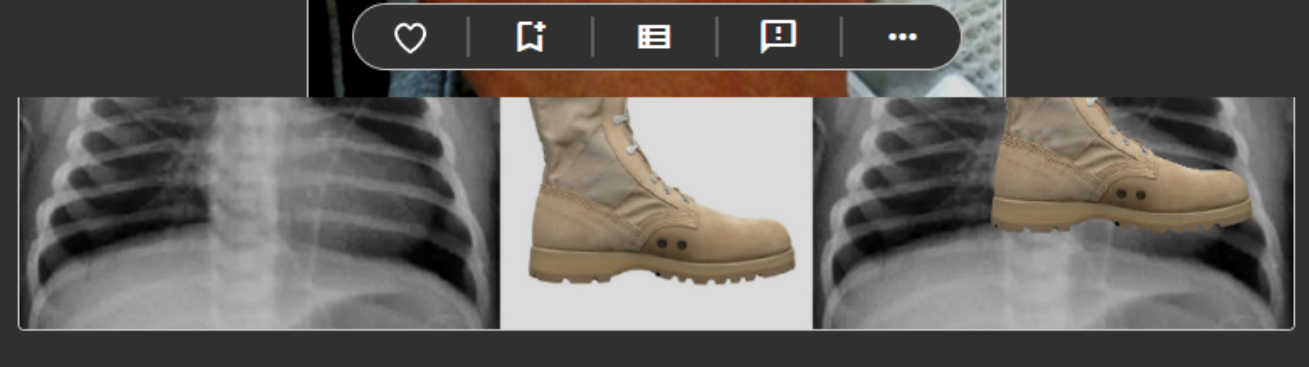
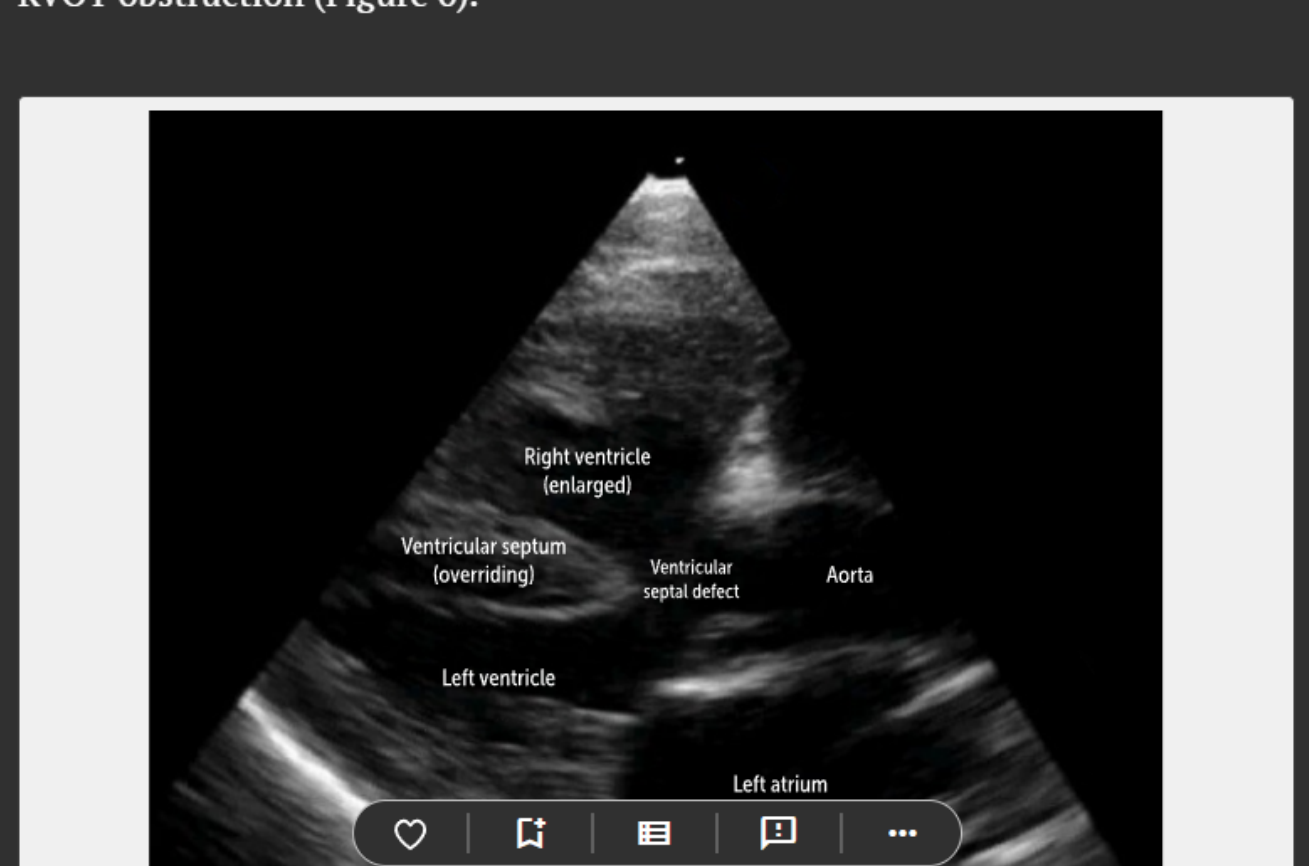


Figure 5 Chest x-ray in tetralogy of Fallot showing boot-shaped heart

Echocardiography. Echocardiography is the definitive diagnostic test for the assessment of TOF. It shows the VSD and can also be used to evaluate the RVOT obstruction (Figure 6).


Right ventricle (enlarged)
Ventricular septum (overriding)
Ventricular septal defect
Aorta
Left ventricle
Left atrium

complications.

Prognosis. Patients with TOF generally do not do very well without surgery. Approximately 50% of patients without treatment do not survive past age 3 years.

What Is Tricuspid Atresia? (LO#1)

As its name suggests, tricuspid atresia refers to the absence of a functional tricuspid valve. This means no direct connection between the right atrium (RA) and RV. In fetal circulation, the foramen ovale allows blood to flow from the right atrium to the left atrium, but unless the foramen ovale remains patent or another atrial septal defect develops, this anomaly is incompatible with life. Additionally, unless another cardiac defect permits left-to-right shunting past the nonfunctional tricuspid valve (like a ventricular septal defect or patent ductus arteriosus), blood will have no way to reach the lungs and will quickly be fatal after birth.

Like TOF, this is a right-sided obstruction and is a **cyanotic** CHD with a **right-to-left shunt**. Therefore, poorly oxygenated blood circulates in the systemic circulation.

TA is less common than TOF, with an estimated incidence of 1 in 10,000 live births.

Clinical Presentation

As stated, blood must be able to shunt from the right atrium to the left atrium to be compatible with ght shunt must be present that allows blood from the left ventricle to supply the pulmonary

What variable defines the early presentation of patients with TA?

Diagnosis

TA is diagnosed by prenatal screening, physical exam, and imaging studies. Most patients are now diagnosed by using fetal echocardiography, as early as 11 weeks.

How are most patients with TA diagnosed?

Physical Examination. Most neonates with TA are cyanotic at birth. Those

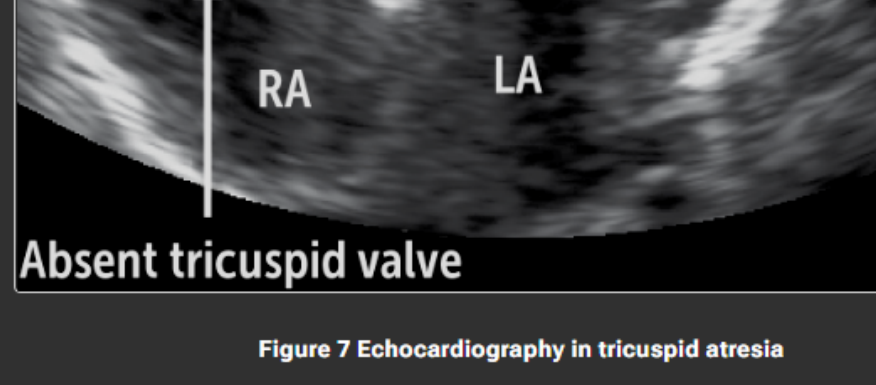
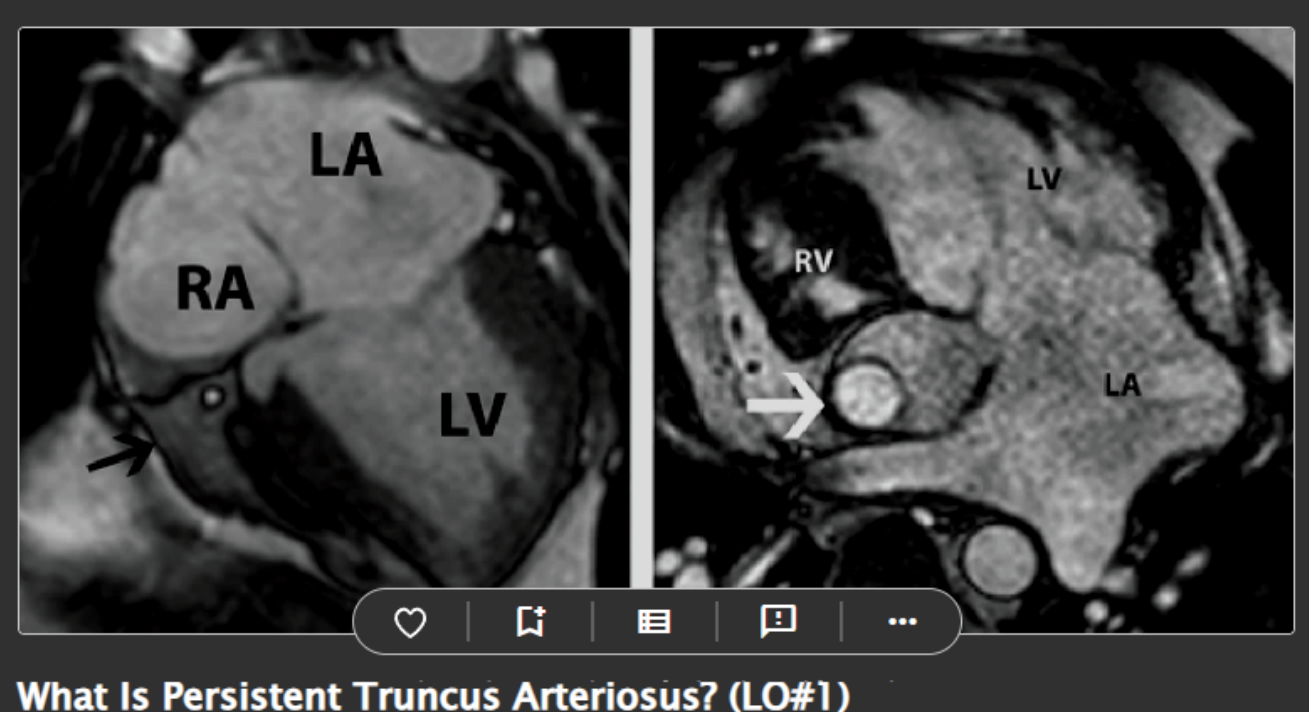


Figure 7 Echocardiography in tricuspid atresia

Color flow Doppler also shows the lack of blood flow through the region where the tricuspid valve should be.

CT and MR Angiography. These will show the absence of tricuspid valve, RA enlargement, and RV hypoplasia (Figure 8).



What Is Persistent Truncus Arteriosus? (LO#1)

A **persistent truncus arteriosus (PTA)** is the failure of the development of the **truncoconal (aorticopulmonary) septum**, an embryologic structure that normally divides to give rise to the pulmonary valve separate from the aortic valve, and the pulmonary trunk separate from the aorta. In infants with persistent PTA, that division is abnormal. Blood is pumped from the heart through a single truncal valve into a single **truncal artery**, which then splits into the aorta and the pulmonary arteries (Figure 9).

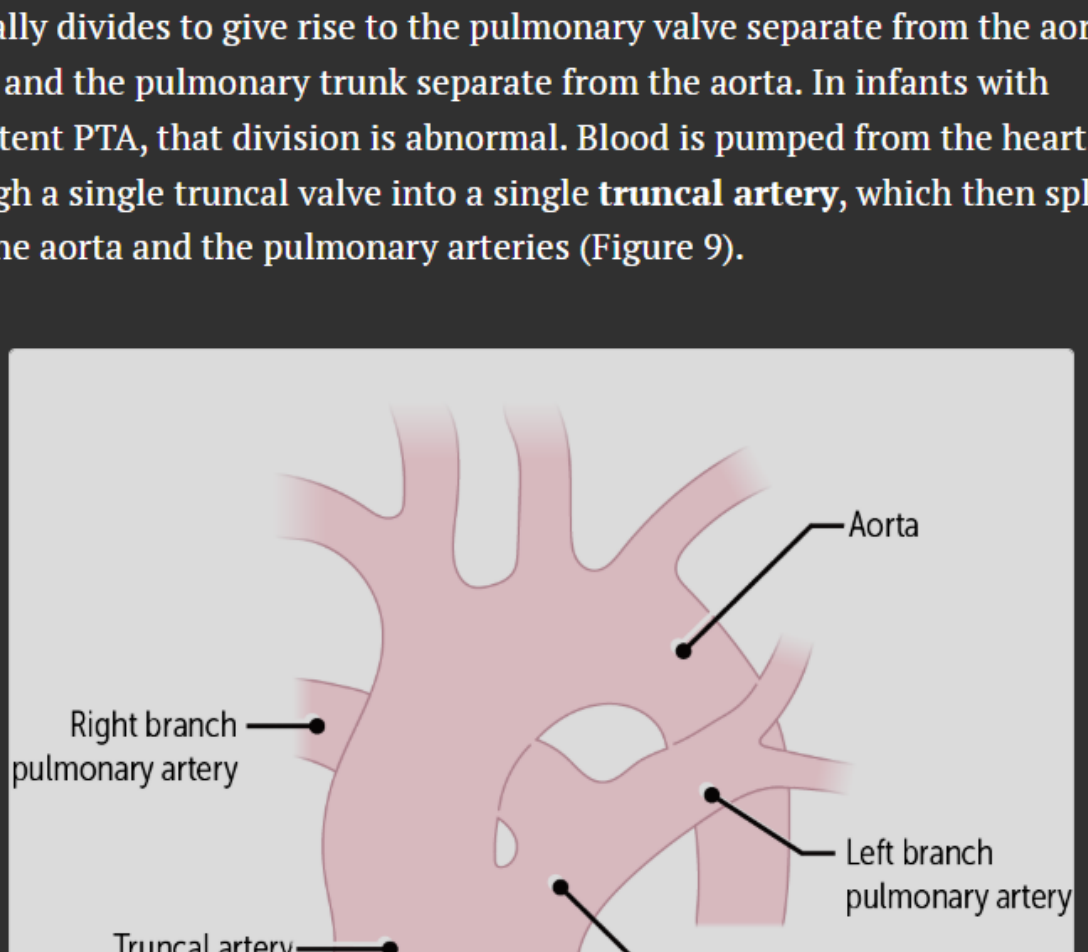


Figure 9 Persistent truncus arteriosus

presence of truncal valve regurgitation, a diastolic murmur may be heard. Genetic testing is required to diagnose 22q11 mutations to look for associated DiGeorge syndrome or velocardiofacial syndrome. Imaging for diagnosis includes fetal echocardiography or a postnatal computed tomography (CT) angiogram.

Management. Management of this disorder is done in two stages, medical stabilization followed by surgical repair:

- Diuretics to reduce pulmonary edema preoperatively--importantly, supplementary oxygen may be harmful for these patients.
- Surgical repair for VSD closure to ensure that blood is not being ejected into the truncal root. Aortic arch anomalies also need corrective surgery.

If left untreated, only 15% of patients survive at 1 year, whereas 90% of treated patients survive 10 years, and some even longer.

CLINICAL CORRELATION

Tetralogy of Fallot and PTA both result from failure of neural crest cell migration. Hirschsprung disease is another such disease. In this case, the cells do not reach the end of the colon to form the ganglion cells or enteric nervous plexuses. The colon cannot function as normal, and megacolon develops.

What Is Transposition of the Great Arteries? (LO#1)

In TGA, also known as transposition of the great vessels or D-transposition of the great vessels (D s d rotation), the

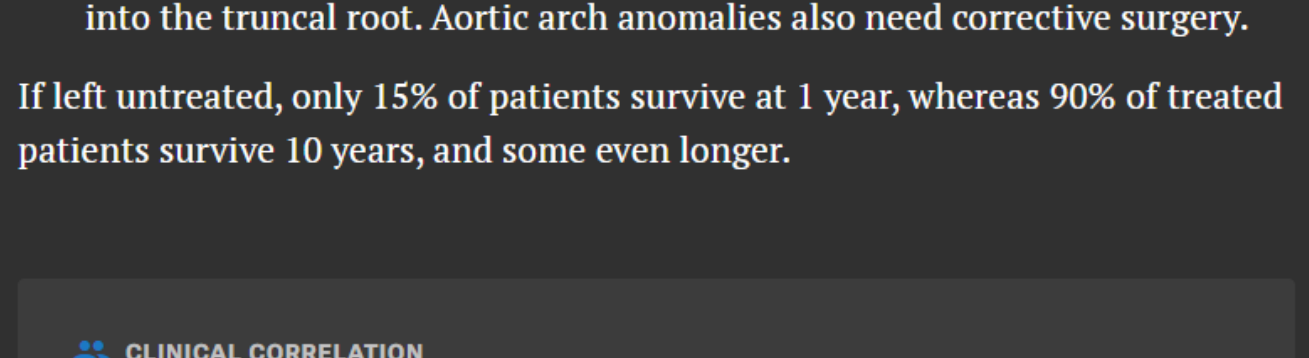
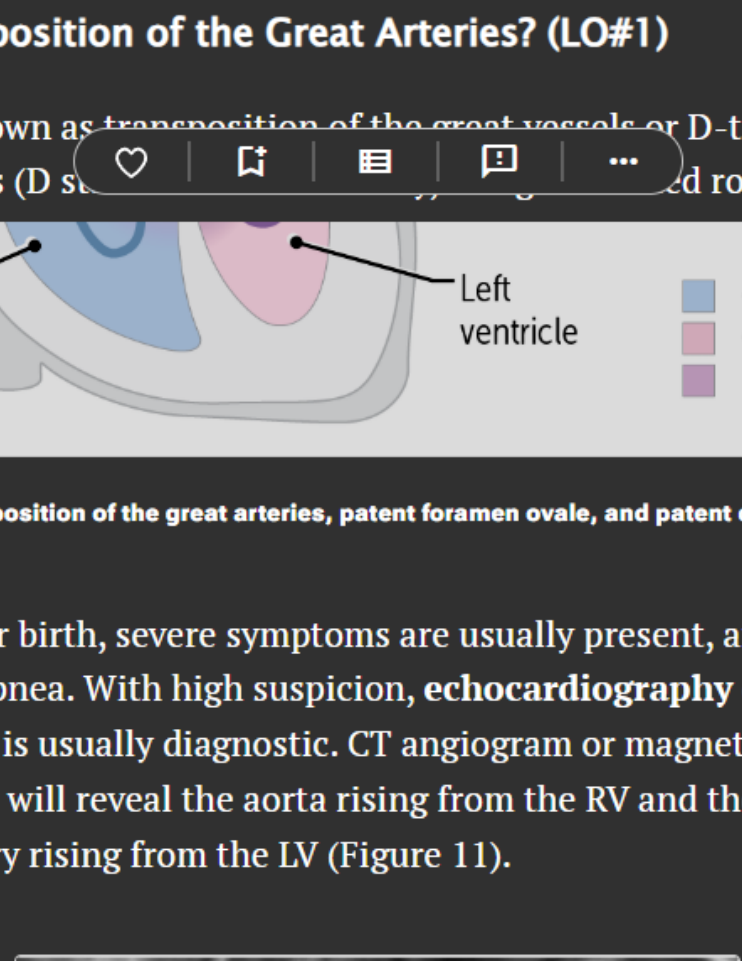


Figure 10 Transposition of the great arteries, patent foramen ovale, and patent ductus arteriosus

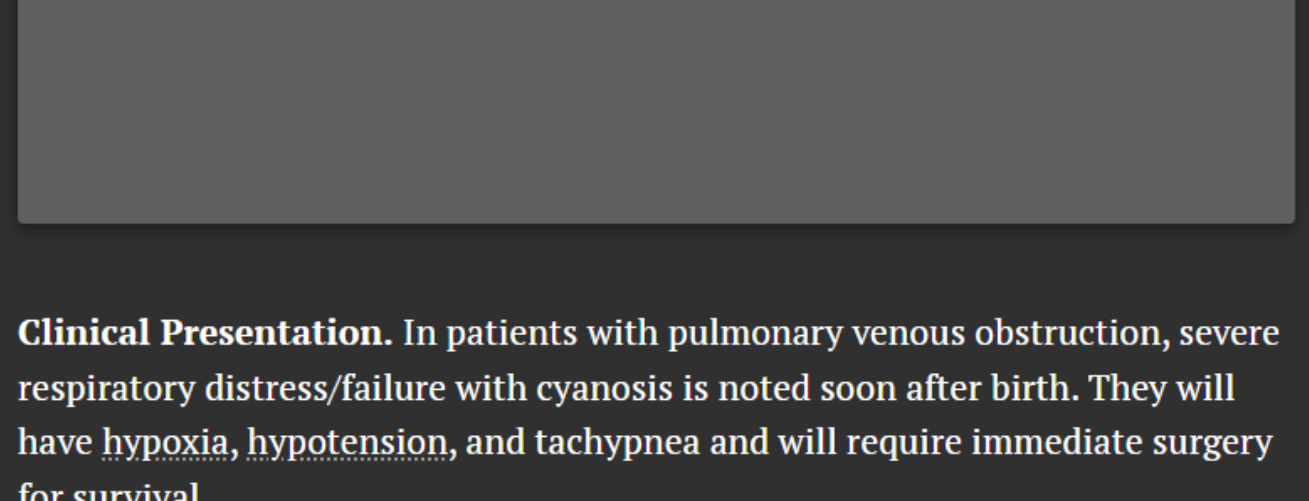
Diagnosis. After birth, severe symptoms are usually present, and newborns may have tachypnea. With high suspicion, **echocardiography** done in the early neonatal period is usually diagnostic. CT angiogram or magnetic resonance (MR) angiogram will reveal the aorta rising from the RV and the main pulmonary artery rising from the LV (Figure 11).



Clinical Presentation. In patients with pulmonary venous obstruction, severe respiratory distress/failure with cyanosis is noted soon after birth. They will have hypoxia, hypotension, and tachypnea and will require immediate surgery for survival.

However, when pulmonary venous obstruction is not present, the degree of cyanosis can be variable, starting slowly over the first few days of life. Patients can have dyspnea, feeding difficulties, and failure to thrive. Later, these patients develop RV hypertrophy and RV failure.

Pathophysiology. Oxygenated blood from the pulmonary veins returns to the right side of the heart, where it mixes with oxygen-poor blood (Figure 12).



of presence of obstruction.

With surgical correction, 80% to 90% of patients live longer than 10 years. However, without treatment, 80% of patients die within the first year of birth.

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back to PB, what is your diagnosis? How do you explain the condition and its treatment to his parents?

The echocardiogram shows the four classic anatomic defects of tetralogy of Fallot: pulmonic stenosis, right ventricular hypertrophy, overriding aorta, and ventricular septal defect. You sit down with PB's parents, who are anxious about their son's condition. You explain to them that TOF is a heart defect that affects the normal oxygenation and flow of blood. As a result, blood with a low oxygen level is pumped out of the heart to the tissues and organs. You tell them that this condition can and should be surgically treated, and most patients who have the surgery generally do very well.

Summary

- Cyanotic congenital heart defects (CHDs) are characterized by low blood oxygen content due to right-to-left shunts, causing deoxygenated blood to mix with oxygenated blood in systemic circulation.
- Tetralogy of Fallot (TOF) is a common cyanotic CHD involving four defects: pulmonary valve stenosis, right ventricular hypertrophy, ventricular septal defect, and an ov saturation.

Hide Explanation

The correct answer is single S2 (E). Normally, the S2 sound comes from the aortic valve and the pulmonic valve. Depending on the intrathoracic pressure and blood return, sometimes the S2 is split, which is physiologic. In patients with tetralogy of Fallot, the pulmonic valve is damaged, so it doesn't contribute to the S2 sound. Diastolic murmur best heard at the right upper sternal border (A) corresponds to aortic regurgitation. Fixed splitting of S2 (B) can be caused by an atrial septal defect. Machine-like rumbling heard throughout auscultation (C) corresponds to a patent ductus arteriosus. An S4 (D) heart sound implies a forceful atrial contraction, which may be associated with left ventricular hypertrophy.

2. A 6-month-old patient is brought to the emergency department by her father, who says that the baby turns blue when she cries. When you examine the child, she is irritable, starts crying, and turns blue. What is the best next step in management?

A. ● Let the parent hold the child with her knees toward her chest
B. ○ Prescribe her morphine and send her home
C. ○ Schedule her for elective intracardiac repair
D. ○ Schedule her for elective shunt surgery
E. ○ Start her on oxygen and admit for observation

causing it to become hypoplastic. The blood now must pass from the right atrium (C) into the left atrium (A), then to the left ventricle (B), and out to the systemic circulation. To reach the pulmonary circulation, the blood must enter the RV via a ventricular septal defect or directly reach the vessels via a patent ductus arteriosus.

How would you rate this Brick?

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